BrainImation: A Browser-Based Teaching Environment for Learning Creative Coding with Live EEG, Optical PPG/fNIRS, and Head Motion Sensors

Kyle E. Mathewson[a,b,*], **Tamari Shalamberidze**[b], **Claire J. Scavuzzo**[a]

[a]Department of Psychology, Faculty of Science, P-217 Biological Sciences Building, University of Alberta, Edmonton, AB, T6G 2E9, Canada
[b]Neuroscience and Mental Health Institute, Faculty of Medicine and Dentistry, University of Alberta, Edmonton, AB, T6G 2E1, Canada

Corresponding Author:
Kyle E. Mathewson, Ph.D.
Department of Psychology, Faculty of Science
P-217 Biological Sciences Building
University of Alberta, Edmonton, AB, T6G 2E9, Canada
Email: `kyle.mathewson@ualberta.ca` (K.E. Mathewson), `shalambe@ualberta.ca` (T. Shalamberidze), `scavuzzo@ualberta.ca` (C.J. Scavuzzo)
Live Tool: https://kylemath.github.io/Brainimation/
Source Repository: https://github.com/kylemath/Brainimation

Target Journal: *Behavior Research Methods* (Original Article / Software Tool)

Keywords (6): electroencephalography (EEG); functional near-infrared spectroscopy (fNIRS); photoplethysmography (PPG); brain–computer interface (BCI); event-related potential (ERP); neuroscience education

Declarations

Open Practices Statement: All source code (`muse-browser.js`, `index.html`, Python `bleak` hardware diagnostics in `tools/`), the live browser environment (https://kylemath.github.io/Brainimation/), all 234 catalogue sketches and thumbnails in `examples/`, simulation benchmark generators, and the automated manuscript verification suite (`verify_manuscript_analyses.py`) are openly available without restriction at https://github.com/kylemath/Brainimation and https://github.com/kylemath/eegeduPaper. The classroom evaluation analyzed de-identified archival student code rather than a preregistered hypothesis test.

Ethics and SAGER Statement: Classroom code artifacts from `PSYCH 403` were analyzed in de-identified form in accordance with procedures approved by the University of Alberta Research Ethics Board (Protocol `Pro00050069`). Because software source files were analyzed in de-identified archival form without demographic linkage, participant sex and gender dimensions were not recorded and do not affect the deterministic signal-processing benchmarks.

Data and Code Availability: The live BrainImation environment is available at https://kylemath.github.io/Brainimation/. Source code and analysis scripts are archived at https://github.com/kylemath/Brainimation and https://github.com/kylemath/eegeduPaper.

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CRediT Authorship Contribution Statement: Kyle E. Mathewson: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Tamari Shalamberidze:** Methodology, Software, Validation, Investigation, Data curation, Writing – review & editing. **Claire J. Scavuzzo:** Conceptualization, Methodology, Investigation, Supervision, Writing – review & editing.

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